ARXIV:2504.02882

DiaTool-DPO

Multi-Turn Direct Preference Optimization for Tool-Augmented LLMs

Enhancing dialogue flow via DPO with automatic paired trajectory construction

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Hallucinated Tool Calls

SFT models invoke tools even when required arguments are missing or incomplete.

Result: incorrect API calls with garbage parameter values.

FAILURE MODE

Missing Follow-ups

Fail to ask clarifying questions when user queries lack required slot information.

Result: silent assumption of default or hallucinated values.

FAILURE MODE

No Rejection

Cannot reject out-of-scope requests when no suitable tool is available.

Result: forced invocation of irrelevant or wrong tools.

FAILURE MODE

Insight: DPO with paired trajectories addresses these gaps without requiring human preference labels.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

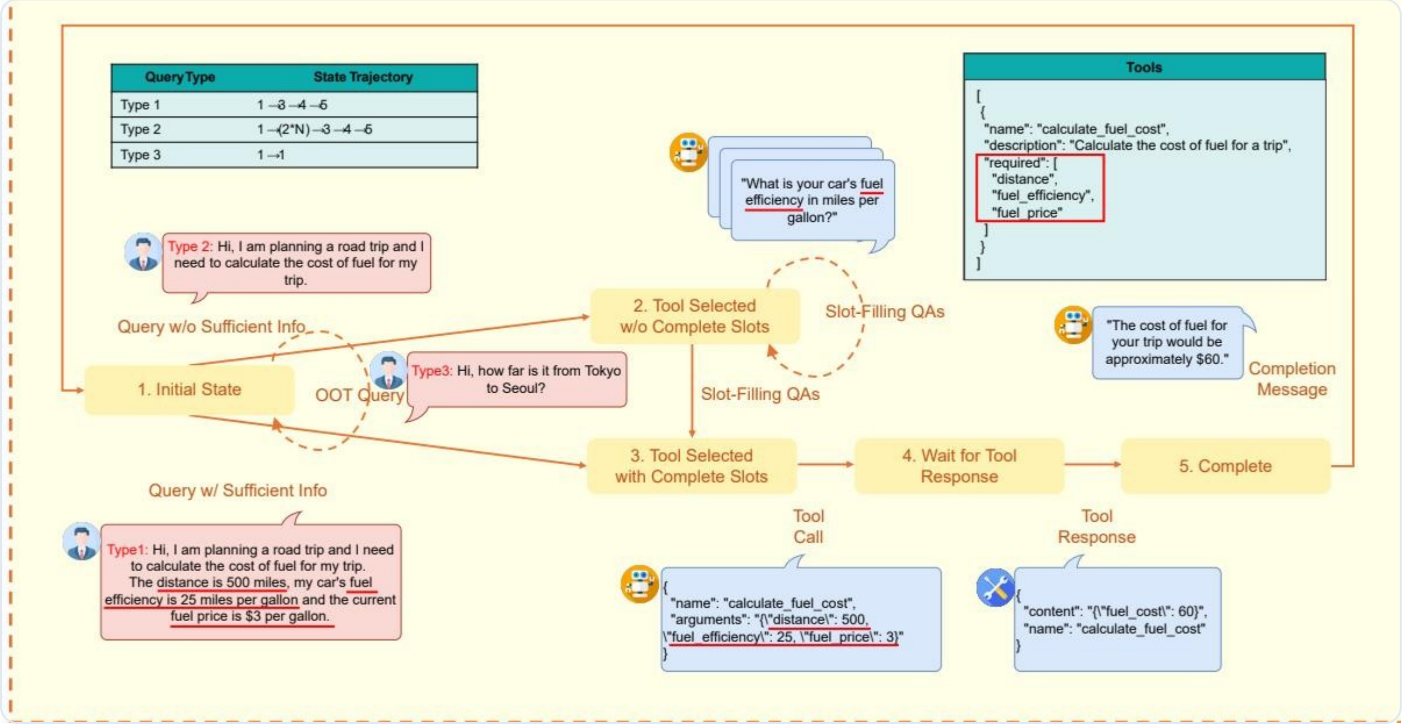


Figure: Visualization of five internal states and state trajectories for three query types (source: arXiv:2504.02882)

Five Internal States

1. **Initial State**
   No dialogue history.
   Out-of-scope → reject.

2. **Tool Selected w/o Slots**
   Missing arguments.
   Slot-filling QA occurs.

3. **Tool Selected w/ Slots**
   All arguments known.
   Ready to call tool.

4. **Wait for Tool Response**
   Tool call executed.
   Awaiting results.

5. **Complete**
   Results received.
   Completion generated.

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Description	Example
Type 1 Sufficient Info	1 → 3 → 4 → 5	Query contains all required arguments. No slot-filling needed. Direct tool call.	*"Book a flight from NYC to LAX on May 10"* (all args present)
Type 2 Incomplete Info	1 → (2×N) → 3 → 4 → 5	Query lacks some arguments. Slot-filling QA repeats N times until all required fields gathered.	*"Book a flight to LAX"* (missing origin, date) → N follow-up questions
Type 3 Out-of-Scope	1 → 1	Requested functionality is unavailable. Tool call must be rejected and user informed.	*"What's the weather?"* (no weather tool available) → reject with explanation

These three trajectories form the basis for automatic preference pair construction — every deviation from the correct trajectory becomes a "rejecte

Automatic Paired Trajectory Construction Without Human Labels

Query	Chosen Trajectory	Rejected Trajectory	Learning Lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2×M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090 / 562	Easy, Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 2	1→(2×N)→3→4→5	1→(2×M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 2	—	1→1	Tool call accept	2,089 / 562	Easy, Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2×N)→3→4	Tool call reject	562	Hard

Total: 11,153+ preference pairs — all generated automatically from the MDP structure without human annotation.

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	**0.833**	0.609	0.575	0.614
Prop.-8B	SFT-only	**0.900**	0.916	0.694	**0.913**	0.870	0.856
Prop.-8B	**SFT + DiaTool-DPO**	0.886	**0.929**	**0.833**	0.826	**0.884**	**0.868**
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	**0.771**	0.817	0.750	**0.826**	**0.790**	0.791
Prop.-3.1B	**SFT + DiaTool-DPO**	0.743	**0.871**	**0.833**	**0.826**	0.765	**0.818**
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	**0.957**	0.639	0.826	0.844	0.816
LLaMA-3-8B	**SFT + DiaTool-DPO**	**0.857**	0.929	**0.917**	**0.913**	**0.905**	**0.904**

Green bold= best value per metric column | Highlighted row = best Macro Average for each model

Source: FunctionChat-Bench evaluation (arXiv:2504.02882)

PERFORMANCE DEEP-DIVE

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-D

Slot Accuracy

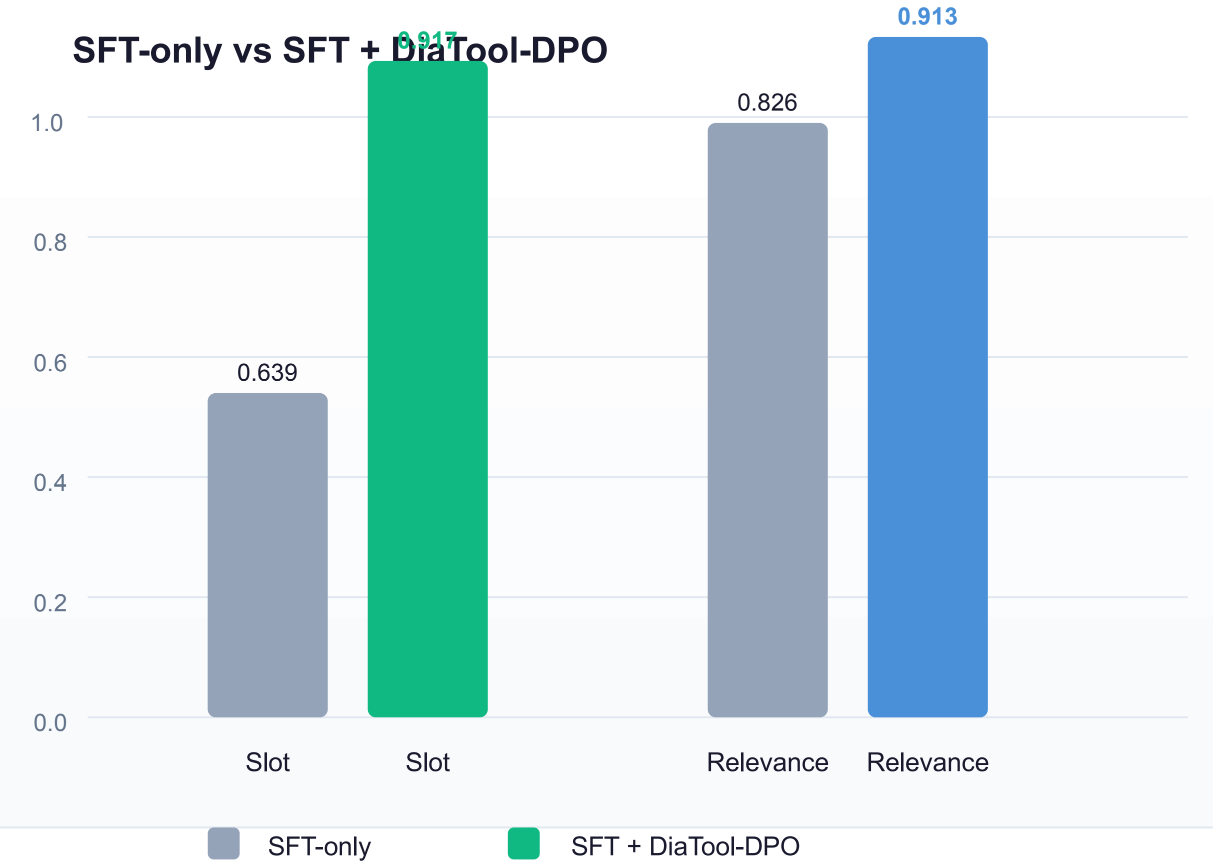
0.639 → **0.917**

+43.5% improvement

Relevance (Tool Rejection)

0.826 → **0.913**

+10.5% improvement



Near GPT-4o performance: 94.8% in information gathering, 91% in tool call rejection — achieved on an 8B open-source model.

KEY TAKEAWAYS

RL-Based Dialogue Control for Production TA-LLMs

1. **MDP formulation enables systematic pairing**
   5 states + 3 query types = automatic generation of chosen/rejected trajectories

2. **Automatic dataset construction eliminates human labeling**
   11K+ preference pairs generated from trajectory rules alone

3. **DPO complements SFT — it does not replace it**
   DiaTool-DPO-only performs poorly; combined SFT + DPO achieves best results

4. **Slot-filling and tool rejection see the largest gains**
   +43.5% Slot accuracy, +10.5% Relevance on LLaMA-3-8B

5. **Language-agnostic and production-ready**
   Generalizes beyond Korean; opens path to GPT-4o-level tool use on smaller models